ERIC XU

exu23@ucsc.edu | ericxu1070.github.io | github.com/ericxu1070 | linkedin.com/in/exu06

EDUCATION

University of California, Santa Cruz Santa Cruz, CA
M.S. in Scientific Computing and Applied Mathematics *2026 – 2027*
B.S. in Applied Mathematics, Highest Honors in the Major | GPA 3.77/4.00 | Dean's Honors (4 quarters) *2024 – 2026*

- **Probability, statistics and finance:** Probability and Statistics, Classical and Bayesian Inference, Financial Mathematics
- **Numerical and computational:** Scientific Machine Learning (grad), Applied Dynamical Systems (grad), Scientific Computing, Computational Methods, GPU Programming, Intro to PDEs

RESEARCH EXPERIENCE

TACS Lab, UC Santa Cruz Santa Cruz, CA
Researcher, B.S. and M.S. (Advisor: Prof. Ashesh Chattopadhyay) *Sep 2025 – Present*

- Built an AI ensemble forecasting pipeline (GenCast + FourCastNet3 with rare-event resampling) for 3-week heat wave and cold outbreak forecasts, validated against the NOAA CFSv2 baseline on closest-member error and critical success index.
- Building a diffusion model that downscales ERA5 (1°) reanalysis to HRRR (3 km) temperature and wind, trained on paired 2015–2025 snapshots.

Vayuh.ai
ML Researcher (part-time) *Jul 2026 – Present*

- Run multi-GPU training and inference experiments for subseasonal extreme-weather forecasting on H100 nodes (SLURM) and NCAR Derecho (PBS).

TEACHING EXPERIENCE

Baskin School of Engineering, UC Santa Cruz Santa Cruz, CA
Teaching Assistant: Differential Calculus for Economics; Ordinary Differential Equations *Sep 2026 – Present*

- Lead weekly discussion sections and office hours; grade assignments and help run course logistics.

Tutor and Reader: Linear Algebra, Discrete Mathematics, Calculus II *Sep 2025 – Jun 2026*

- 3 quarters of tutoring experience and 1 quarter as a grader.

PROJECTS

4D-Var Data Assimilation with a Learned Inverse Operator | *PyTorch, L-BFGS* Apr – May 2026

- Reimplemented a Google Research 4D-Var method with a learned CNN inverse observation operator in PyTorch; cut state-estimation RMSE by **38–75%** vs standard 4D-Var on Lorenz-96.

Cross-Species Transfer Learning with DeepSEA | *PyTorch, scikit-learn* Jun 2026

- Fine-tuned the human-trained DeepSEA CNN on **~1M** mouse ENCODE DNA windows; beat training from scratch on **115 of 133** chromatin features (median AUROC 0.877 vs 0.865).
- Reached best validation loss in **5 epochs vs 12** from scratch; showed zero-shot transfer is near chance (AUROC 0.50), so fine-tuning is required.

Council of Sumer | *Python, FastAPI, LLM agents, OpenTelemetry* Oct 2026

- Built a hackathon system (team of 3) where **5 LLM agents** with conflicting goals run as separate processes, debate over AGNTCY SLIM, and vote, with runoff and human tie-break rules.
- Wrote the deliberation engine, tracing, and benchmark: across 16 seeded games the council kept **73%** of cities vs **56%** for a single LLM and starved **27% fewer** people.

SKILLS

Mathematics: Probability and statistics, numerical methods, linear algebra, ODEs and PDEs, dynamical systems
Languages and Tools: Python, C++, R, MATLAB, Bash, RISC-V assembly; Linux, Git, SSH
ML, Scientific Computing and HPC: PyTorch, JAX, NumPy, scikit-learn, Triton (GPU kernels), SLURM, PBS